

Networks and Security

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Basic Firewall Configuration in Cisco Packet Tracer

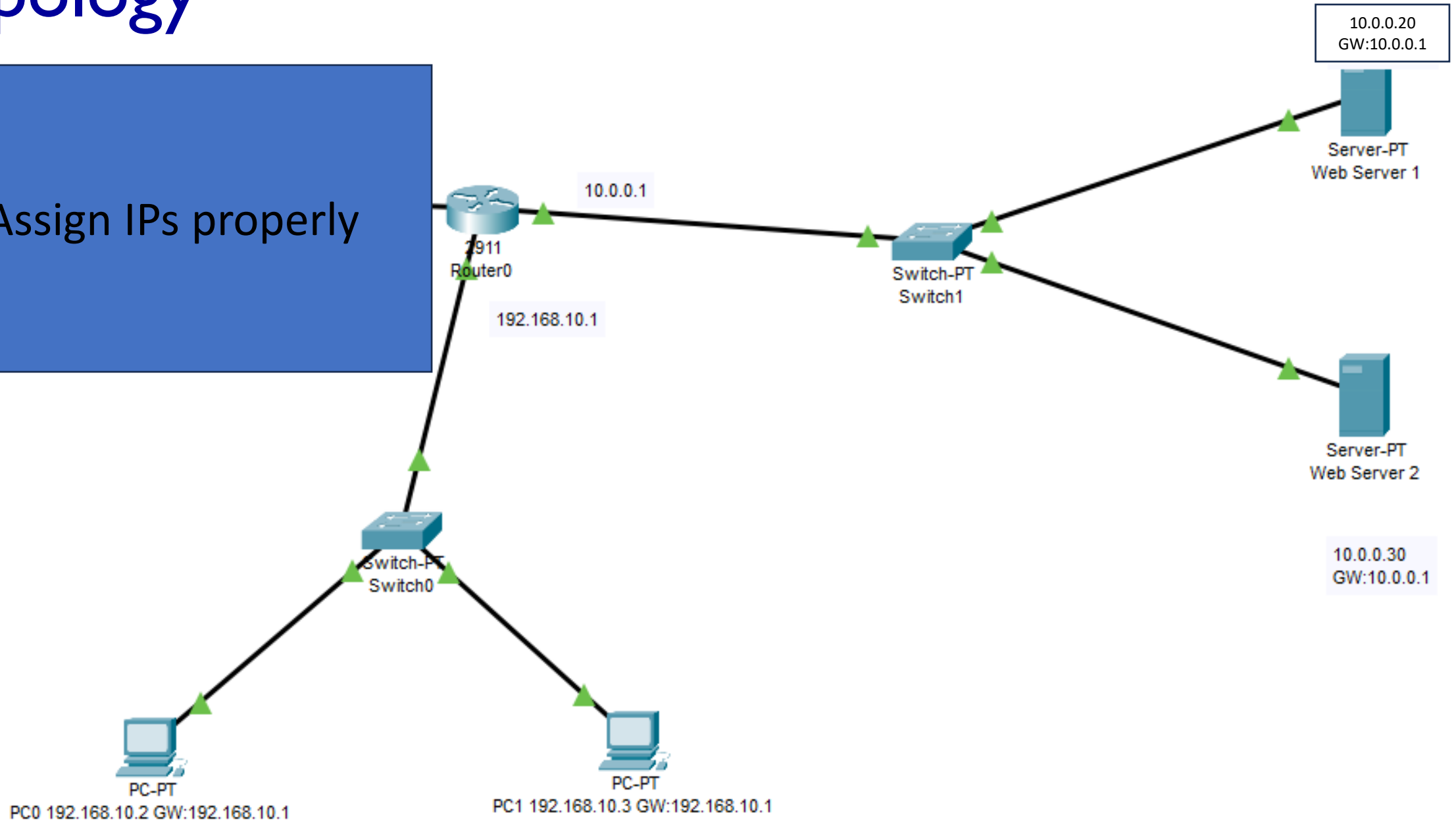
Introduction

First, open the Cisco packet tracer desktop and select the devices given below:

Device	Model Name	Quantity
PC	PC	2
Router	2911	1
switch	2960-Switch	2
Server	Server	2

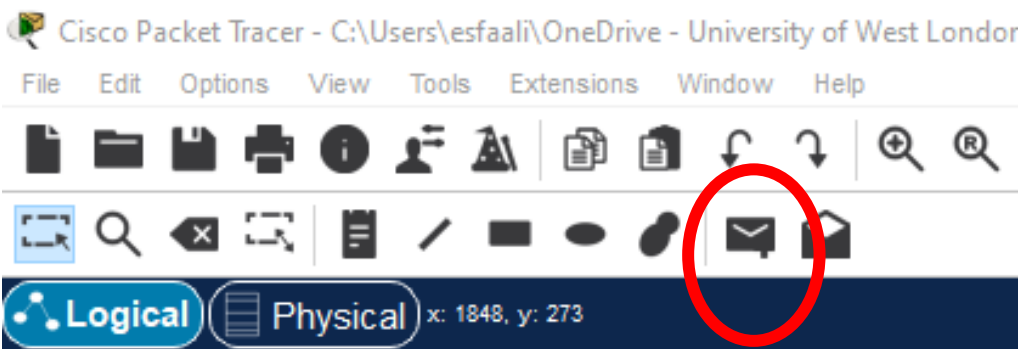
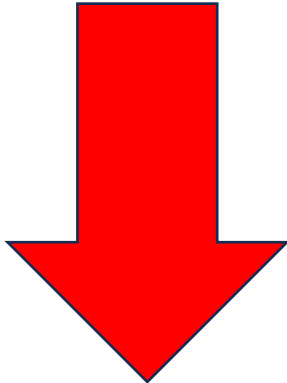
Topology

Assign IPs properly



Check the status

Click on  and drop it at PC0 or PC1 to see all devices are connected!

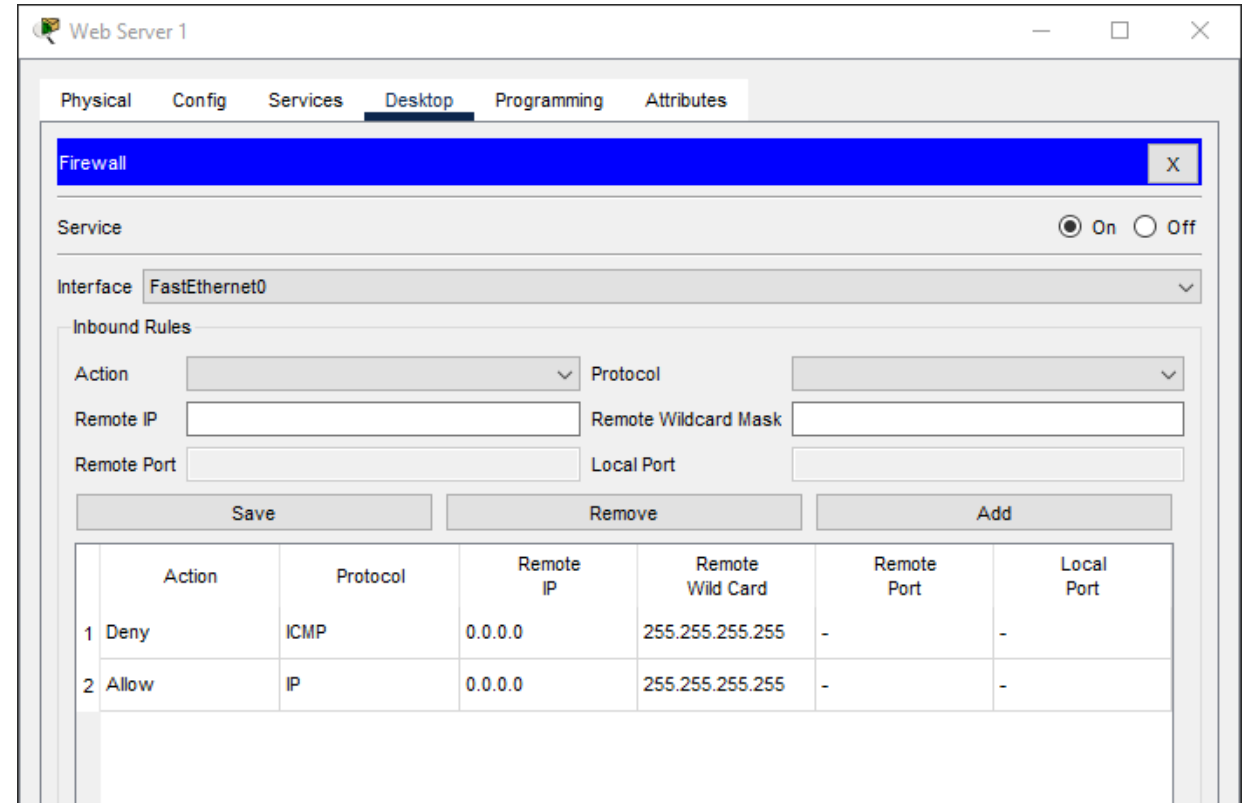


Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	
	Successful	PC0	Web Serv...	ICMP		0.000	N	1	(edit)	
	Successful	PC1	Web Serv...	ICMP		0.000	N	2	(edit)	
	Successful	Web ...	Web Serv...	ICMP		0.000	N	3	(edit)	

Firewall Configuration

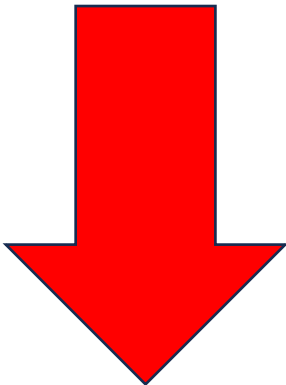
In Web Server 1, we want to:

- Deny ICMP: Blocks ping from PCs
- Allow HTTP: PCs still can access to the http service (visit the website server)
- Click on Web server 1 then go to the desktop, then click on firewall IPv4, and turn on the services.
- First, choose deny from action, then choose ICMP from protocol and set remote IP to 0.0.0.0 and Remote wildcard mask to 255.255.255.255, finally add it.
- Second, choose allow from action, then choose IP from protocol and set remote IP to 0.0.0.0 and Remote wildcard mask to 255.255.255.255, and add it.



Check the status- Deny ICMP

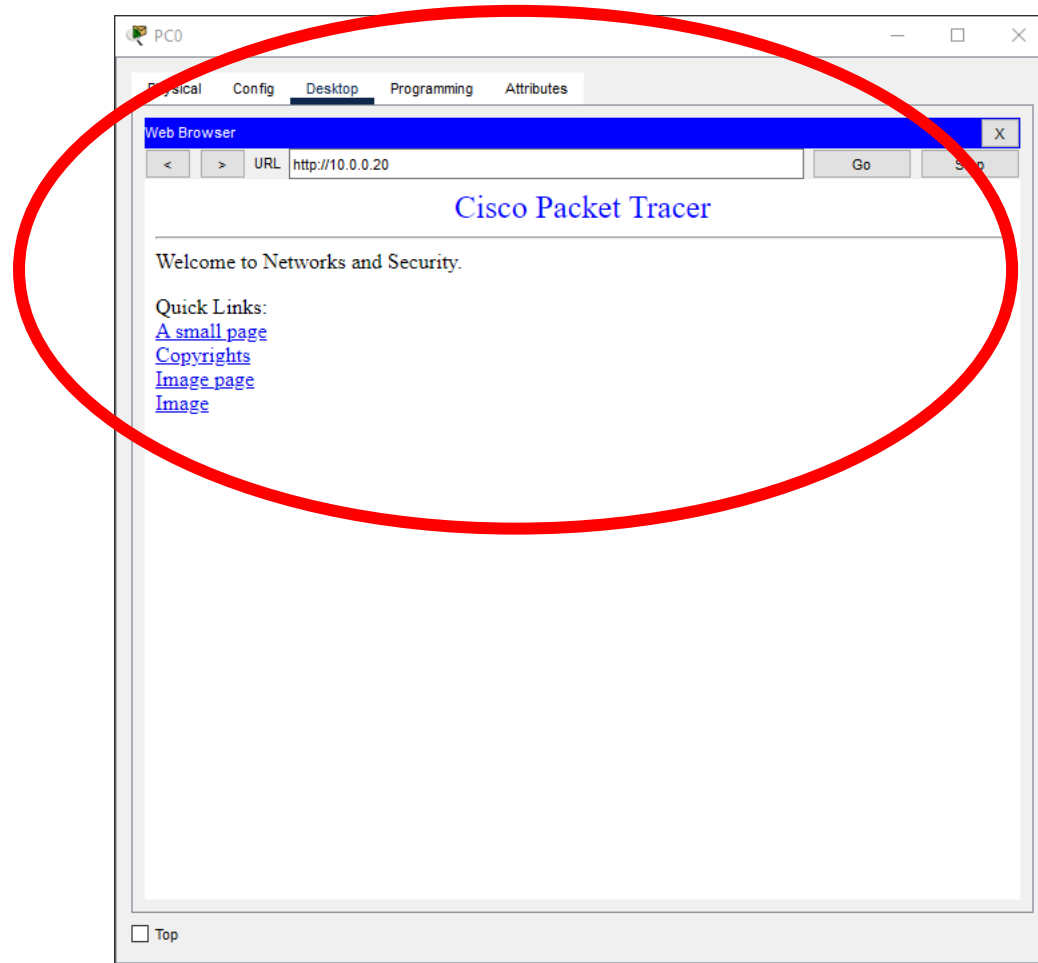
Click on  and drop it at PC0 or PC1 to see all devices are connected!



re	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delet
	Failed	PC0 19...	Web Server 1	ICMP		0.000	N	0	(edit)	
	Failed	PC1 19...	Web Server 1	ICMP		0.000	N	1	(edit)	
	Successful	PC0 19...	Web Server 2	ICMP		0.000	N	2	(edit)	
	Successful	PC1 19...	Web Server 2	ICMP		0.000	N	3	(edit)	

Check the status- Allow HTTP

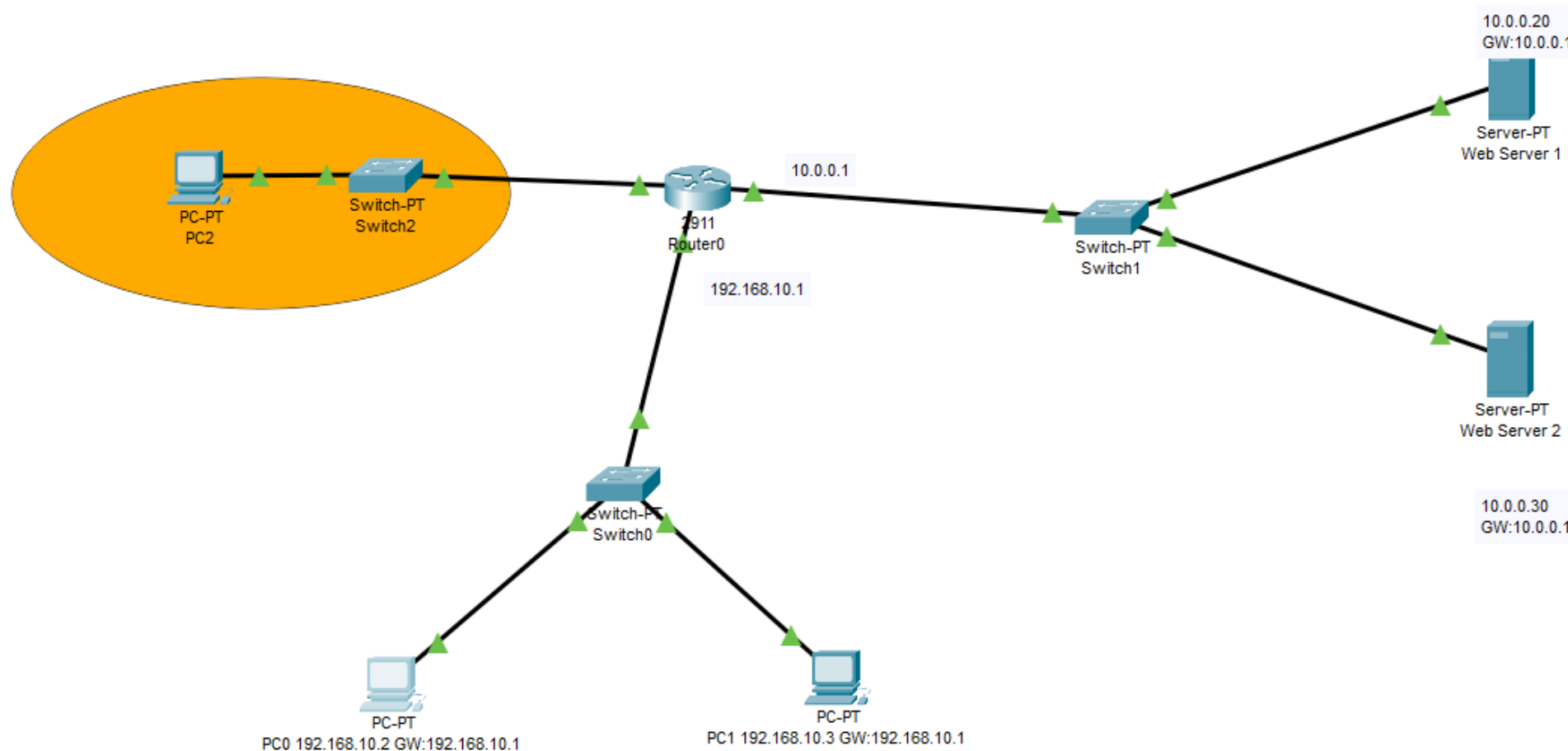
Go to PC0, click on Web Browser, and then type the address of Web server 1 (10.0.0.20), and then press Go



Exercise

Objectives

Add a network, as shown below, to your existing network to enable access to Web Server 1.



Hints

- You may need to first allocate IP addresses to the new PC and the router connected to it.
- You may also need to update the firewall policy on Web Server 1.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Dele
	Successful	PC2	Web Server 1	ICMP		0.000	N	0	(edit)	

This is what you need to show us